

Module 08: Planning in Robotics — Autonomous Mission Planning

Learning Objectives

1. Define **autonomous mission planning** as the capacity to decompose high-level goals into executable action sequences over extended time horizons.
2. Analyze how **hierarchical task planning, contingency planning, and replanning** implement Active Inference at the strategic level.
3. Apply mission planning frameworks to autonomous systems operating in dynamic, uncertain environments.

Introduction

Mission planning is the highest level of the Active Inference hierarchy in autonomous robots — determining *what* to accomplish over hours, days, or weeks of operation. While trajectory planning asks "how do I get from A to B?", mission planning asks "where should I go, in what order, and what should I do at each location?" This is strategic cognition for autonomous systems.

Key Concepts

1. Hierarchical Task Decomposition

Complex missions are decomposed into hierarchical task structures:

- **Mission level:** "Survey the disaster zone and locate survivors" (hours)
- **Task level:** "Search Grid Section C" → "Navigate to Section C" + "Execute sweep pattern" + "Report findings" (minutes)
- **Action level:** "Navigate to waypoint 7" → trajectory, obstacle avoidance, localization (seconds)

Each level performs Active Inference at its appropriate timescale — high levels set goals for low levels, low levels report completion and observations back up. The mission planner evaluates Expected Free Energy at the mission level while delegating execution details to lower levels.

2. Contingency Planning

Robust mission plans include **contingencies** — pre-computed responses to anticipated failures:

- "If battery drops below 15%, abort current task and return to charging station"
- "If communication link is lost, complete current task then return to base"
- "If survivor is detected, interrupt sweep pattern and switch to close-inspection mode"

Each contingency is a conditional policy with a trigger condition (prediction error threshold) and a response policy. Pre-computing contingencies allows rapid response without the computational cost of replanning from scratch.

3. Online Replanning

When the actual situation diverges too far from the planned scenario, the autonomous agent must **replan**:

- **Trigger:** Rising free energy that cannot be resolved by contingency policies — the plan's predictions no longer match reality
- **Scope:** Replanning can be local (adjust the current task) or global (restructure the entire mission)
- **Cost:** Replanning is computationally expensive — the agent should replan as rarely as possible, using contingencies and reactive adjustments to handle minor deviations

4. Multi-Objective Mission Optimization

Real missions involve multiple competing objectives:

- **Coverage vs. depth:** Should the surveyor cover the whole area quickly or examine suspicious regions thoroughly?

- **Safety vs. speed:** Should the delivery drone take the short route over a populated area or the long route over open terrain?
- **Information vs. action:** Should the robot gather more data before acting or act on current (imperfect) estimates?

Expected Free Energy naturally handles multi-objective trade-offs through the pragmatic-epistemic decomposition — coverage/speed/action serve pragmatic value; depth/safety/information serve epistemic value.

5. Temporal Constraints and Scheduling

Mission planning often involves **temporal constraints**:

- Task deadlines ("deliver package by 3 PM")
- Task dependencies ("inspect before repairing")
- Resource windows ("solar-powered robot must complete outdoor tasks during daylight")

These constraints structure the EFE evaluation — policies that violate temporal constraints receive infinite pragmatic cost (they cannot achieve preferred observations within the time window).

Applications

- **Disaster response drone swarm:** A swarm of search-and-rescue drones plans a coordinated survey of a disaster area — hierarchically decomposing the search zone into sectors, with each drone planning its sector sweep, contingency policies for battery depletion and communication loss, and replanning triggers for survivor detection.
- **Long-duration ocean survey:** An autonomous underwater glider plans a month-long oceanographic survey — optimizing sampling locations to maximize scientific information gain (epistemic EFE) while satisfying battery constraints, ocean current predictions, and scheduled surface communications for data upload.

Conclusion

Mission planning is Active Inference at the strategic level — hierarchical decomposition, contingency planning, replanning, multi-objective optimization, and temporal scheduling all implement Expected Free Energy minimization over extended horizons. This completes the Autonomous Agents unit and the Robotics domain course.